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BUSINESS INTELLIGENCE

PROJECT PROPOSAL

Title:

Retail Strategy: The Impact of GDP and Urbanization on European Sales

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Problem Background

Business Context

This project focuses on a retail company that operates across the European market and sells different types of products through both online and offline channels. The company already has internal sales data which does provide a clear view of customers transaction history. However, this data alone does not explain the external factors that influence customer buying behaviour.

To gain deeper insights, this project combines internal sales data with external data such as nominal GDP which represents the countries' economic performance and the demographic factor, specifically urban population percentage. These datasets are integrated using Alteryx Designer to build a Business Intelligence solution. The goal is to understand how a country's economic strength and level of urbanisation affect customer shopping preferences.

Importance of the Problem

In today's competitive global market, retail companies face high risks when expanding into new countries. If they rely only on past sales data, decisions may be based on simple assumptions only rather than the real socio-economic market. For example, investing heavily in physical stores in a country with high urbanisation but low GDP growth may result in poor returns compared to focusing on online sales. This issue shows the gap between raw data and meaningful insights. Solving this problem is important because data-driven decisions can help companies improve profitability and operational efficiency..

Expect Impact

The implementation of this Business Intelligence project is expected to provide several important benefits for the organization. First, it will support strategic channel optimization by helping the organization decide whether to focus more on e-commerce or physical retail stores based on a country's level of urbanisation. Secondly, it will enable targeted economic growth by identifying high-potential markets through the relationship between sales performance and GDP per capita. Finally, this project will provide decision support by developing an interactive dashboard using Microsoft Power BI that allows executives to easily understand complex socio-economic trends. This will help the organization move from reactive reporting to a more proactive and data-driven strategy.

Project Objectives

1. To use Alteryx to combine our internal European sales records with external data on country wealth (GDP) and city living (Urbanization) into one single, clean dataset.
2. To identify if there is a clear connection between a country's GDP per capita or Urbanization rate and whether customers choose to shop Online or Offline.
3. To build an interactive dashboard using Microsoft Power BI that allows users to easily visualize sales trends and filter results by country, wealth and living environment.
4. To provide data-driven recommendations on which European countries should receive more investment in physical retail stores and which countries should focus on expanding digital e-commerce.

Dataset Description

To achieve the project objectives, three distinct structured datasets have been selected. These datasets will be integrated to provide a holistic view of how economic and demographic factors influence retail performance.

Dataset 1: Europe Sales Records

This dataset contains retail sales transaction records from multiple countries in Europe. It includes information related to sales channels, item types, revenue, profit, and order details. The dataset is useful for analyzing sales performance, comparing online and offline channels, and identifying regional sales trends.

- **Source:**
Kaggle – [Europe Sales Records Dataset](#)
- **Type of Data:**
Structured data (CSV format)
- **Dataset Size:**
Approximately 1,000 rows and 14 columns
- **Key Variables / Features:**

Variable	Data Type	Description	Example Value
Order ID	Integer	Unique transaction identifier	478051030
Country	String	Country where sales occurred	San Marino

Item Type	String	Product category sold	Office Supplies, Beverages
Sales Channel	String	Online or Offline sales platform	Offline, Online
Order Priority	String	Priority level of the order	M, H
Order Date	Date	Date when the order was placed	9/12/2011
Ship Date	Date	Date when the order was shipped	9/29/2011
Units Sold	Integer	Quantity of products sold	4778
Unit Price	Float/Decimal	Price of products sold	47.45
Total Revenue	Float/Decimal	KPI analysis	226716.1
Total Profit	Float/Decimal	Profitability analysis	74823.48

Dataset 2: Global GDP Explorer 2025 (World Bank UN Data)

This dataset contains economic indicators of countries worldwide, including GDP performance, GDP growth, and GDP per capita. It will be used to examine whether a country's economic condition influences retail sales performance and customer purchasing behavior.

- **Source:**
Kaggle – [Global GDP Explorer 2025 Dataset](#)
- **Type of Data:**
Structured data (CSV format)
- **Dataset Size:**
Approximately 200 rows and 8 columns
- **Key Variables / Features:**

Variable	Data Type	Description	Example Value
Country	String	Official country name	United States
Nominal GDP	String	Total nominal GDP in USD	\$27,720,700,000,000
GDP Growth (%)	String	Annual GDP growth rate (%)	2.89%

Population 2023	Integer	Estimated population for 2023	343477335
GDP per Capita	String	Average income per person (USD)	\$80,706
Share of World GDP	String	Percentage contribution to global GDP	26.11%

Dataset 3: Countries of the World 2023

This dataset provides demographics, social, and development-related information for countries worldwide. It includes urbanization, population, CPI, and wage indicators which can help explain customer purchasing environments and market conditions.

- **Source:**
Kaggle –[Global Country Information Dataset 2023](#)
- **Type of Data:**
Structured data (CSV format)
- **Dataset Size:**

Approximately 195 rows and 35 columns.

- **Key Variables / Features:**

Variable	Data Type	Description	Example Value
Country	String	Country name	Afghanistan
Urban_population	Integer	Percentage of people living in urban areas	9,797,273
Population	Integer	Total population of the country	38,041,754
CPI	Float/Decimal	Consumer Price Index	149.9
Minimum Wage	String	Minimum wage level in local currency.	\$0.43
Density	Float/Decimal	Population density measured in persons per square kilometer	60
Land Area	Integer	Total land area of the country in square kilometers.	652,230

Life Expectancy	Float/Decimal	Average number of years a newborn is expected to live	64.5
Fertility Rate	Float/Decimal	Average number of children born to a woman during her lifetime.	4.47

Proposed BI Approach

To convert raw socio-indicators and sales data into strategic insights, this project implements a structured BI pipeline. The architecture split into a backend ETL (Extract, Transform, Load) process and a frontend visualization dashboard layer. By utilizing Alteryx Designer for backend data engineering and Tableau or Power BI for frontend analytics, we create a seamless pipeline from raw data ingestion to executive decision-making.

Alteryx Designer

Alteryx serves as the primary engine for data integration and preparation. It is responsible for building the ETL pipeline through the following methodology:

1. **Data Cleaning and Type Correction** : Before merging, Data Cleaning Tool will use on each individual dataset to handle null value and trim the leading whitespaces. The Select Tool will then be used to manually cast “Urban Population%” and “GDP” from strings into numeric types to remove any string-based formatting that would prevent mathematical analysis.
2. **Data Integration**: Once individual dataset are standardized, Join Tool will be used to merge the datasets together. By utilizing "Country" as the common primary key, the process generates a unified master file that links internal sales performance directly to the specific economic and demographic profile of each nation
3. **Data Transformation**: To prepare the data for visualization, we will reshape and refine the master file.
 - **Dimensional reduction**: Select Tool will be used to deselect redundant or irrelevant columns. Since merging three large datasets creates an excess of variables, we will limit the final output to essential fields only to improve dashboard performance.
 - **Data Re-Typing**: The Select Tool will be used to cast string-based economic figures into numeric formats to enable mathematical analysis.
 - **KPI Derivation**: New business metrics will be derived by using Formula Tool to provide a deeper strategic context. “Profit per Urban Resident” will be calculated to normalize sales performance against the actual urban population of each market as the primary KPI.

4. Output: The finalized workflow will export a cleaned master dataset in .csv format that is designed for high-performance analysis in the visualization layer.

Tableau or Power BI

The frontend visualization platforms will translate the processed data into an interactive dashboard that focuses on correlating wealth and urbanization with shopping behavior. The analysis plan focuses on three core objectives:

- Wealth and Profit Correlation: Examine the relationship between GDP per capita and total profit to determine whether wealthier countries generate higher sales value.
- Urbanization and Channel Preference: Identify whether highly urbanized countries show a stronger preference for online shopping compared to offline retail.
- Market Efficiency Assessment: Evaluate “Profit per Urban Resident” to identify the most efficient markets for retail expansion.

Below are the proposed visualizations for this project.

Visual Form	Specific Analysis Plan	Strategic Business value
Geospatial Map	Mapping total sales volume across European countries	Identifies geographic "hotspots" and regions where market presence is lacking.
Scatter Plot	Correlating GDP per Capita with Total Profit	Determines if national wealth is a primary driver for higher-margin transactions.
Grouped Bar Chart	Comparing Online vs. offline profit across Item Types	Informs supply chain strategy by identifying which products thrive in digital vs. physical channels.
Heatmap	Analyzing relationship between Urbanization and Sales Channel usage	Visualizes whether higher population density corresponds to a preference for e-commerce.
Line Chart	Tracking Monthly Sales Trends across the years	Identifies seasonal peak windows for proactive stock and marketing planning.